DeepStream parallel multi-model inference

> Applicable environment: JetPack 7.2.1, Ubuntu 24.04, DeepStream 9.1, Jetson Orin NX. Complete the DeepStream environment setup first and confirm that the basic USB or CSI camera cases can run.

> The factory system already provides the DeepStream, the models, the TensorRT engines, the course startup entry and the compatible configuration that this case needs. In normal use you start directly from section 3 and choose the input source to run; the deployment commands of this section are only for reference when the environment is missing, has to be restored or has to be redeployed.

1. Factory environment and first deployment reference

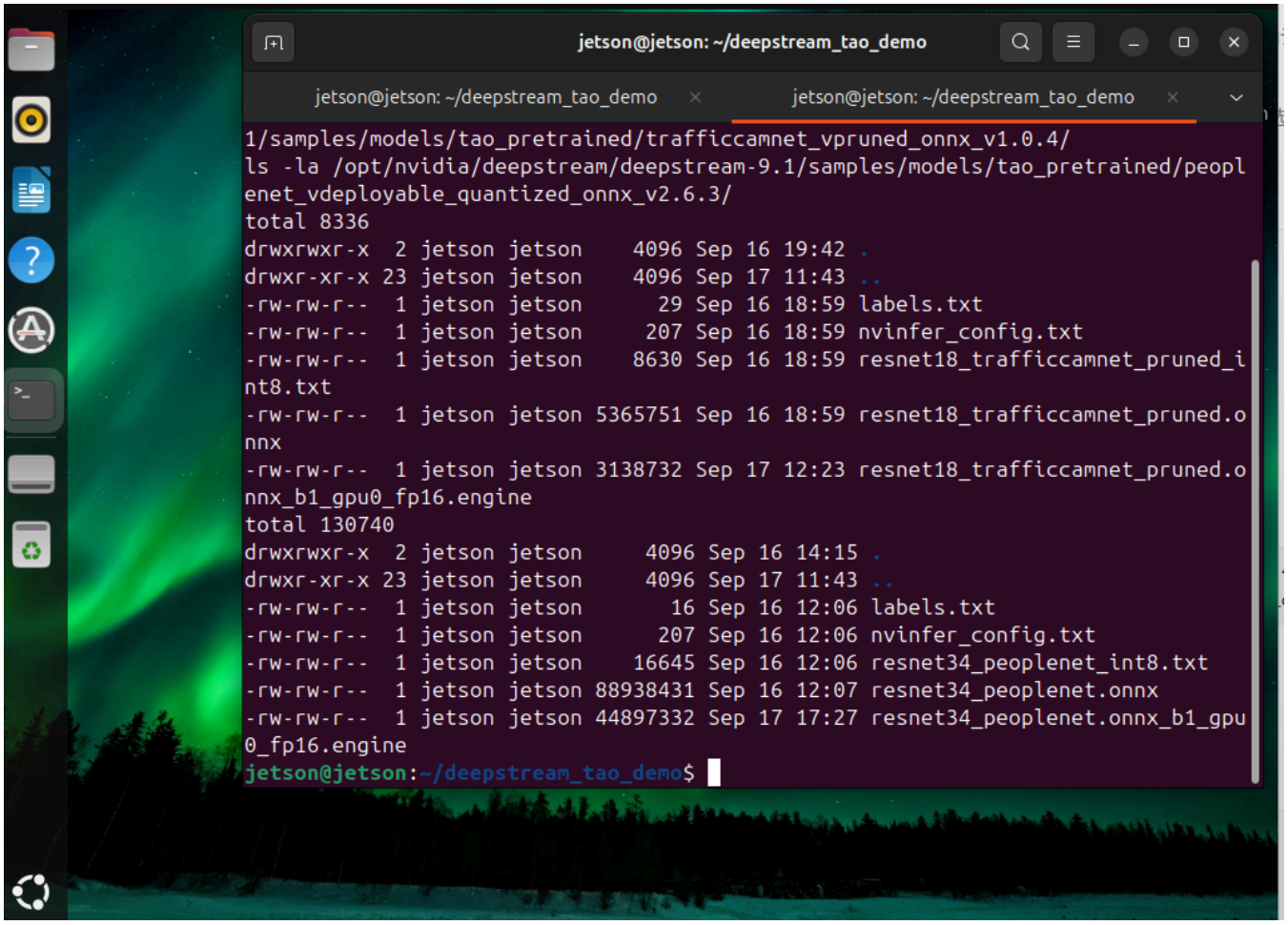
The factory system already contains:

- The three startup scripts for the video, the USB and the CSI under `~/deepstream_tao_demo/`.
- The TAO pretrained models and the TensorRT engines required by TrafficCamNet and PeopleNet.
- The plugins required by the GStreamer pipeline and the `nvdsmetamux` metadata merging plugin.

Therefore you must not run the commands of this section again; only when the environment is missing, has to be restored or you deploy it yourself, follow the steps below in this order.

1.1 Confirm that the models are in place

```
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/trafficcamnet_vpruned_onnx_v1.0.4/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/peoplenet_vdeployable_quantized_onnx_v2.6.3/
```



1.2 Build the common TAO parser (only needed the first time)

```
cd ~/deepstream_tao_apps/post_processor
export CUDA_VER=13.2
make clean
make
ls -la libnvds_infercustomparser_tao.so
```

2. Case goal

This case derives four display tiles from the same video frame or the same camera frame: the original image, TrafficCamNet, PeopleNet and the merged metadata result of the two models.

The final window is a 2×2 four-tile layout:

Tile	Content that it displays
Top left	The original image
Top right	TrafficCamNet + Tracker + NVDS Analytics
Bottom left	PeopleNet
Bottom right	The merged metadata result of TrafficCamNet and PeopleNet

The four tiles are composited into the same window by the `compositor`; the bottom right merges the metadata of the two inference branches with `nvdsmetamux`.

For USB/CSI, the camera is captured only once; DeepStream copies the same frame internally with a `tee` and does not open the camera again.

3. Script startup: choose the input source

The current unified startup script is located in `~/deepstream_tao_demo/`:

```
cd ~/deepstream_tao_demo
```

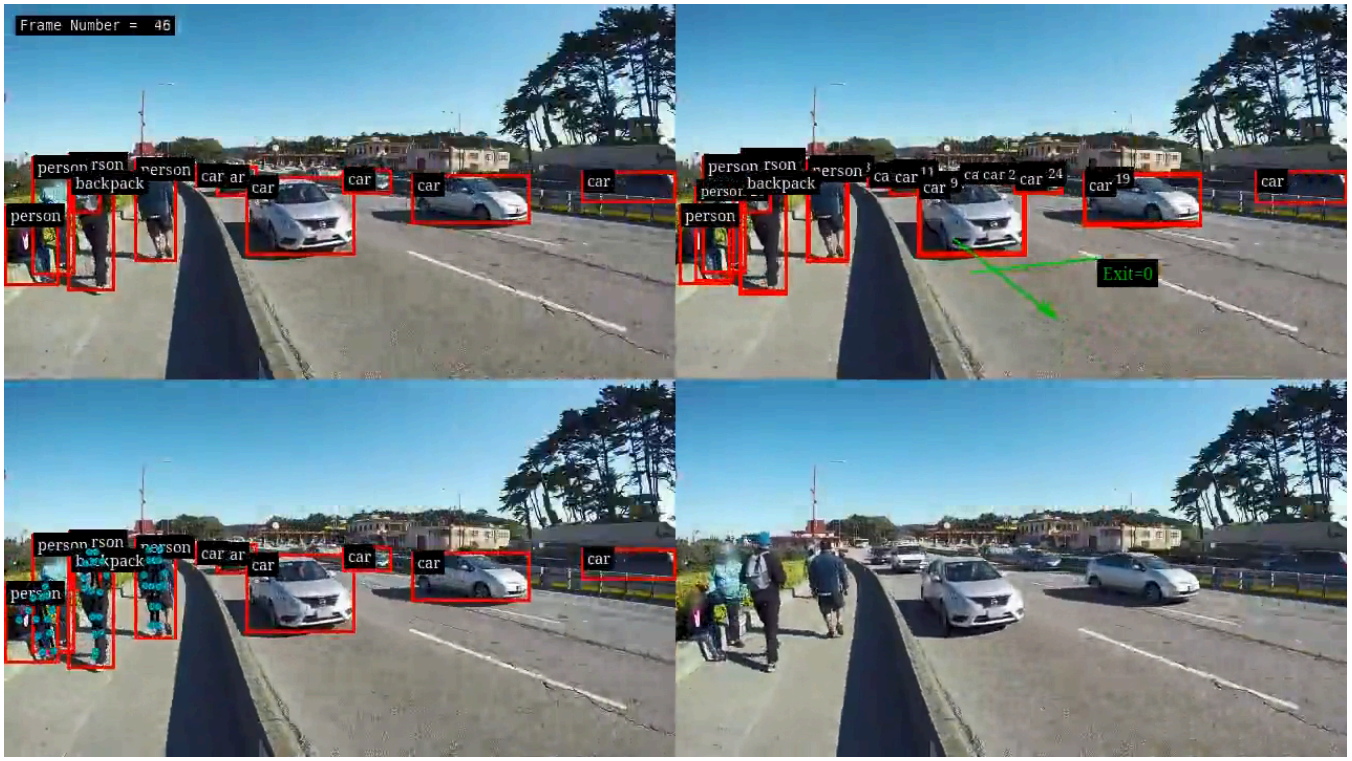
Input source	Command	Applicable scenario
Local video	`./run_4grid.sh`	The default sample_720p.h264, to verify the overlay effect of the two models
USB camera	`./run_parallel_gst_usb.sh`	Real-time input of a UVC/V4L2 USB camera
CSI camera	`./run_parallel_gst_csi.sh`	Real-time input of a MIPI CSI camera

All three entries display a single window in real time and do not write a video file.

4. Local video verification

For the first run it is recommended to start with the local video and confirm that the two models, the four tiles and the display environment are correct:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_4grid.sh
```



The default video: `/opt/nvidia/deepstream/deepstream/samples/streams/sample_720p.h264`

Replace the video:

```
./run_parallel_gst.sh /path/to/your_video.mp4
```

/path/to/your_video.mp4 is the path of the video that you want to run the detection on

5. Real-time verification with the CSI camera

Run:

```
cd ~/deepstream_tao_demo
./run_parallel_gst_csi.sh
```

By default it uses the CSI `sensor-id=0`, `wbmode=1` (automatic white balance) and an exposure compensation of 0. When the plugin supports
`sensor-mode`, it requests mode 3; when it does not, the mode is selected by the `1920x1080@30` caps.

6. Real-time verification with the USB camera

Prerequisites (confirm the USB camera node and the supported formats):

```
v4l2-ctl --list-devices
v4l2-ctl -d /dev/video0 --list-formats-ext
```

The script detects the USB camera node automatically through `~/deepstream_tao_demo/find_usb_cam.sh`, so there is no need to specify it manually.

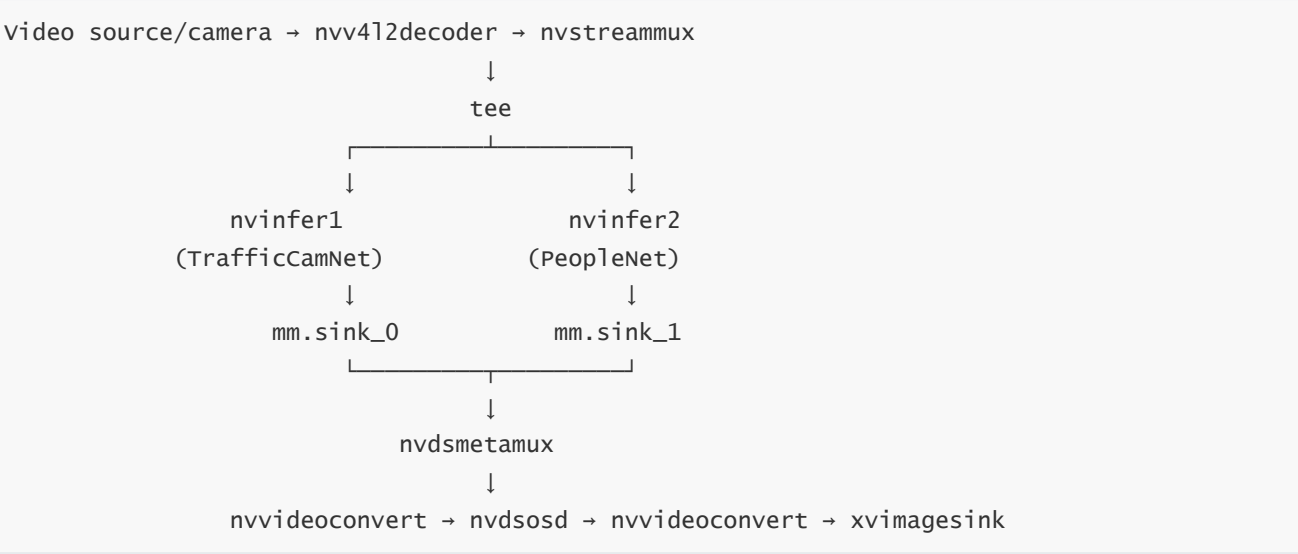
Run:

```
cd ~/deepstream_tao_demo
./run_parallel_gst_usb.sh
```

The USB camera uses `640x480@15fps` by default in the YUYV mode (a USB 2.0 bandwidth limit). If a higher resolution is needed, the camera must support the MJPG mode; the script currently takes the YUYV path by default.

7. Dual-model parallel processing chain

The processing chain is:



- TrafficCamNet detects traffic targets such as vehicles, bicycles and road signs.
- PeopleNet detects person targets such as people, faces and bags.
- `nvdsmetamux` merges the metadata of the two branches into one stream and then sends it to `nvdsosd` to draw the boxes uniformly.

Key configuration (already set up in the script):

- TrafficCamNet uses `gie-unique-id=1`, with the configuration file `pgie_trafficcamnet_config.txt`.
- PeopleNet uses `gie-unique-id=2`, with the configuration file `config_infer_primary_peoplenet_id2.txt` (`batch-size=1`, matching the b1 engine).
- The outputs of the two GIE are fed into `nvdsmetamux` through `queue ! mm.sink_X`.

8. Output and stopping

The real-time window is displayed on the desktop of the Jetson device itself or on the VNC desktop. None of the three scripts writes a video file.

Press `Ctrl+C` once in the terminal where the command runs to stop it normally.

9. Common phenomena

Symptom	Handling
Only one set of boxes is visible and the two sets cannot be told apart	There should be 2 `Load new model` entries in the log (UID 1 + UID 2); the boxes of the same color overlay each other and are hard to distinguish visually, so check the log to confirm
The USB version fails to start	Run `v4l2-ctl --list-formats-ext` and check the device number, the resolution and the frame rate
The CSI cannot be opened	Confirm that `sensor-id=0` corresponds to the actual camera and check the Argus and the camera occupancy; the script stops the pose/OCR relay first
The terminal reports `not-negotiated`	The resolution of the USB camera does not match the caps of the script; change `width/height/framerate` in the script
The picture stutters	The two models are inferred at the same time; the FPS is about 15-20 on the Orin NX, which is normal